

Question Paper

Exam Date & Time: 24-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIRST SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025
SUBJECT: MME 1003/MME_1003 FUNDAMENTALS OF MECHANICAL ENGINEERING

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) 500kg of wet steam at a pressure of 0.6 MPa is generated in a boiler per hour. The temperature of feed water is 20°C and the total amount of heat added in the boiler is 1236MJ/h. The steam from the boiler enters the super heater after a heat loss of 300kJ/kg, where its temperature is to be raised to 200°C. Determine (4)
- (i) Dryness fraction of the steam at the entry point of the super heater?
(ii) Heat absorbed per hour in the super heater?

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 1B) A gas-fired boiler produces superheated steam at 100 bar from feedwater at 45 °C, with a steam flow rate of 12 kg/s. Fuel gas is burned at a rate of 1.5 tons per hour, having a GCV of 46 MJ/kg, and the boiler efficiency is 88%. Find the condition of steam produced. (3)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 1C) Classify boilers based on the following criteria: (3)
- i) Flow of water and hot gases
ii) Circulation of water
iii) Axis of the shell

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 2A) Power is transmitted between two shafts, 4 meters apart, using two pulleys, both rotating in counterclockwise directions, with a velocity ratio of unity. The length of the belt is 11.142m, and the coefficient of friction is 0.3. Determine the size of the two pulleys and the tensions on both sides of the belt drive if the initial tension is 0.75KN. (4)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 2B) A simple gear train as shown in Figure, has four gears namely A, B, C and D. Gear A is driver gear and D gear is driven gear. Gears B and C are intermediate gears. Driving gear A rotates anticlockwise direction with a speed of 300 rpm. The number of teeth on gears A, B, C and D are 30, 90, 60 and 120, respectively. Determine (3)
- (i) the speed of the driven gear, and (ii) the speed of rotations of intermediate gears, (iii) the velocity ratio of gear trains.

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 2C) Explain, with a neat diagram, the arrangement and operation of a pulley system designed to transmit motion to the follower shaft at various speeds. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 3A) Illustrating thermodynamic events explain working of an IC engine that directly admits the charge and operates with a compression ratio of 9:1. (4)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 3B) The following observations were recorded during a test on an IC engine. Net load on the brake drum = 36kg, Cam Shaft Speed= 250 rpm, Fuel consumption= 0.003kg/sec, Diameter of the brake drum = 2m, Calorific value of the fuel= 43900kJ/kg. Determine the Brake Thermal Efficiency and Brake specific fuel consumption. (3)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 3C) Give suitable reasons for the following: (3)
- i) Crank shaft speed is double that of cam shaft speed.
 - ii) Piston rings are an essential component of an internal combustion engine.
 - iii) Four stroke engines require a heavier flywheel as compared to two stroke engines.

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4A) A vapour compression refrigerator uses tetrafluoroethane (R134a) as a refrigerant and operates between 10.1 bar and 2 bar. The vapour is superheated before entering and after exiting the compressor by 7.5°C and 18°C respectively. Compute the following: (4)
- i. Coefficient of performance of the system
 - ii. Mass flow rate of the refrigerant in the system, if the capacity of the system is 14 TR.

- 4B) With a neat sketch, explain the working principle of a domestic refrigerator. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4C) With a neat sketch, explain any three lathe operations. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5A) Explain the various modes of heat transfer with suitable examples. (4)

Course Outcome: "CO5" / Cognitive Level: "Application"

- 5B) State and explain any three features of CNC control system. (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

- 5C) Explain the step-by-step procedure for producing an object using additive manufacturing method. (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 MME_1003:FUNDAMENTALS OF MECHANICAL ENGINEERING

#	Course Outcome	Description	Marks
1	1	CO 1	17
2	3	CO 3	13
3	4	CO 4	6
4	5	CO 5	10
Total			46

* **NA** - CO Not Selected

Event Name	Subject
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#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	1	3(6.52%)
3	Application	7	22(47.83%)
4	Analysis	6	21(45.65%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	14	46 (100%)